

ALEXANDRE LEUCKERT KLEIN

Senior Software Engineer | Full-Stack, Embedded & IoT Systems | Technical Leadership | Startup & Deep Tech

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FIND ME ONLINE

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SUMMARY

Senior Software Engineer with experience building end-to-end products across embedded systems, backend platforms, and customer-facing applications. I've helped startups turn ideas into production-ready systems by working across the entire technology stack—from electronics and firmware to cloud services and modern web applications. My work has directly contributed to the development of core products and supported the successful raise of multiple millions of dollars in investment.

- My experience includes:
- High-performance real-time processing of massive point cloud datasets
 - Embedded firmware and hardware development (ARM, bare-metal, RTOS)
 - VR applications and Unreal Engine plugins in C++
 - ERP, CRM, and POS platforms integrating payments, fiscal compliance, IoT, and business operations using Laravel and PHP
 - Technical leadership, architecture, and cross-functional product development
 - Foundations in offensive security and cyber intelligence

I enjoy solving complex problems, taking ownership of critical systems end-to-end, and building products that bridge hardware, software, and real-world business needs.

EXPERIENCE

Firmware & Hardware Engineer

Alignment Products Inc

- 03/2024 - Present USA
- Designed and implemented firmware for the company's flagship product using C
 - Developed hardware architectures and integrated embedded systems components
 - Participated in product validation and technical decision-making
 - Worked on low-level software interacting directly with hardware peripherals
 - Contributed to the successful development of a commercial product supporting investment fundraising efforts
 - Collaborated across disciplines in a fast-paced startup environment
 - Technologies: C, ARM, RTOS, Firmware Development, Embedded Systems, Hardware Design

Technical Lead / Senior C++ Software Engineer

Miris / Illusio Inc

- 08/2021 - 03/2024 USA
- Architected and developed core components for a high-performance platform capable of real-time processing of billions of 3D points, utilizing C++17/20.
 - Led technical initiatives, mentored team members, and contributed to strategic architectural decisions facilitating project success.
 - Optimized application performance and throughput in critical workloads through profiling and architectural improvements.
 - Built networking and streaming functionality, integrating TCP/IP, Sockets, and Boost.Asio to support real-time data processing.
 - Delivered a key product that enabled multi-million-dollar fundraising, showcasing technological innovation and market potential.
 - Developed integrations for VR environments and plugins for Unreal Engine, expanding platform capabilities.
 - Implemented multithreaded and asynchronous components for enhanced performance using modern C++ technologies.

Software Developer

FishTV

- 09/2018 - 08/2021 Brazil
- Developed and maintained ERP and CRM systems supporting company operations
 - Built software platforms used for championship and event management
 - Designed and implemented REST APIs and backend services
 - Developed production-ready software across the full development lifecycle
 - Enhanced operational efficiency and contributed to business growth through technology initiatives
 - Technologies: C#, Laravel, JavaScript, REST APIs, SQL, Git

KEY ACHIEVEMENTS

- ERP, CRM & POS for Breweries and Pubs**
Increased revenue 12.5x through the development of an integrated platform.
- Real-Time 3D Processing**
Achieved high-performance processing of billions of 3D points per second in real-time applications.
- Successful Fundraising Contributions**
Contributed to fundraising efforts exceeding \$5 million through innovative product developments.
- Cross-Platform Software Efficiency**
Led a team to develop secure cross-platform software, improving efficiency by 30% in production environments.

PROJECTS

ERP, CRM & POS for Breweries and Pubs

- 03/2024 - Present Novo Hamburgo, Brazil
- Developed an integrated ERP, CRM, and POS platform for breweries, supporting operations, inventory, loyalty, production, reporting, fiscal compliance, payments, and IoT brewing. Built with Laravel, Vue.js, and Vuetify; central for daily operations and management.
- Enabled a 12.5x increase in gross revenue while substantially reducing operational losses and waste through the implementation of an integrated ERP, CRM, and POS platform.
 - Significantly reduced inventory losses and waste through real-time stock control, production tracking, and automated consumption reconciliation.
 - Transformed brewery and pub operations by replacing fragmented manual processes with an integrated ERP, CRM, and POS platform that became the single source of truth for the business.
 - Created a loyalty ecosystem converting visitors into repeat customers, enhancing relationships and revenue.

SKILLS

Languages
C++, C, PHP, Python, JavaScript, TypeScript

Build & DevOps
CMake, Git, CI/CD, Bash, PowerShell

Security
Secure software practices, Cybersecurity fundamentals, Offensive Security concepts, Endpoint security interest, Threat awareness

Cloud
Azure, GCP and AWS concepts, Scalable architectures, Distributed systems

Embedded Systems
Firmware development, ARM microcontrollers, RTOS, Hardware integration, Electronic architecture design

Networking
TCP/IP, UDP, Sockets, HTTP/HTTPS, REST APIs, Boost.Asio, Client-server architectures, Streaming systems, OpenSSL, mbedTLS, Certificate-based communication